Project Proposal: Autonomous Vehicle Vision System

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Theme Selection

We have selected **Theme A: Image Processing**. To simulate the preprocessing vision stages of an autonomous vehicle, we will implement an image processing pipeline consisting of three sequential stages:

1. **Grayscale Conversion:** Reduces the 3-channel RGB image to a single channel to reduce computational load.
2. **Gaussian Blur:** A stencil computation to reduce camera noise and smooth the image.
3. **Sobel Edge Detection:** Identifies the boundaries of roads, cars, and obstacles.

The output of one stage will feed directly into the next stage in the pipeline.

Dataset Plan

We plan to use a high-resolution vehicle dashcam dataset - **100K Vehicle Dashcam Image Dataset** on Kaggle (https://www.kaggle.com/datasets/mdfahimbinamin/100k-vehicle-dashcam-image-dataset). We will test our parallel scalability using the following concrete input sizes:

1. **Small Input:** 512 x 512 images - Used strictly for debugging and rapid correctness testing.
2. **Medium Input:** 1920 x 1080 images - Represents our standard, real-world 1080p dashcam workload.
3. **Large Input:** 4K (3840 x 2160) images - Used specifically to stress-test the MPI communication network overhead and ensure larger core counts are utilized effectively.

Goals and Success Criteria

Goal: The program will output processed image files consisting of isolated edges representing the final stage of the pipeline. We will implement Serial, OpenMP, and MPI versions of the code

Verification: We will write a verification script to compare the binary output of our OpenMP and MPI parallel versions against our Serial Baseline. This will prove our parallel code is correct by ensuring the image matrices match bit-for-bit.

Performance: We target a **6x speedup** on the 4K dataset using 8 cores. We will achieve this by ensuring our decomposition stays local to the processor and minimizing unnecessary communication between nodes.